

Injae Jun

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Education

- BS Seoul National University (SNU), Mechanical Engineering** Mar 2020 – Present
- GPA: 3.77/4.0
 - **Coursework:** Mechatronics (A+), Mechanical Product Design (A+)
 - 2 years of absence due to military service

Research Interests

- Soft Robotics (Novel Actuation Mechanisms, Modeling & Control)
- Wearable Robotics & Exoskeletons (Assistive Devices, Prosthetics, Rehabilitation)
- Mechanism Design & Optimization (Bio-inspired Mechanisms, Compliant Mechanisms)

Research Experience

- SOFT ROBOTICS & BIONICS LAB (SRBL)** Jul 2024 – Present
Research Intern – *Advisor: Prof. Yong-Lae Park*
- Developed a compact **thumb opposition wearable** for patients with thenar muscle dysfunction, integrating tendon-driven and rack-pinion actuators with ergonomic hardware design.
 - Fabricated and integrated EGaln-based stretchable sensors for motion sensing.
- Healthcare Robotics Lab (HeRo Lab)** Jan 2025 – Present
Research Intern – *Advisor: Prof. Amy Kyungwon Han*
- Researched twisted string actuators and designed a **novel bistable transmission mechanism** to enhance their force and speed output.
 - Investigated human hand musculature and grasping biomechanics, applying findings to the design and prototyping of a prosthetic hand system.

Projects

- Auto Balancing Case** [Project Link](#) 
- Prototyped a self-balancing [luggage system that reduces wrist load](#) using load-cell sensing, differential actuation, and reinforcement learning control.
 - Validated robustness through PhysX-based simulation and Sim2Real transfer to hardware.
 - Tools Used: Fusion 360, Arduino, IsaacSim, RSL-RL (PPO), Python
- Auxetic Structure Optimization for Crash Box** [Project Link](#) 
- Modeled auxetic unit cells and optimized their geometry using Grid Search, CMA-ES, and FEA to maximize crash energy absorption.
 - Conducted compression and impact tests on 3D-printed TPU prototypes, achieving up to [26.5% improvement](#) in energy absorption.
 - Tools Used: MATLAB, Python, Finite Element Analysis, CMA-ES, 3D Printing
- Reverse Engineering and Improvement of the FCX24 Lemur** [Project Link](#) 
- Analyzed chassis strength with ANSYS, drivetrain/transmission gear ratios, and suspension dynamics of an RC car model.

- Improved design by shifting COM and tuning suspension damping; fabricated a custom damper and experimentally measured damping coefficient.
- Tools Used: ANSYS, Fusion 360, MATLAB, Zaber, Force Gauge

Multivariate Time Series Forecasting using XGBoost & Neural Networks

[Project Link](#)

- Predicted next-day oil temperature on the ETT dataset by redefining the target as daily difference, achieving **top 1%** accuracy among 200 participants.
- Tools Used: Python, XGBoost, PyTorch, Scikit-learn, Pandas, NumPy

Leadership / Extracurricular activities

SNU Computer Study Club (SCSC)

Mar 2020 – Aug 2020

- Participated in Python and C programming study groups; developed a Minesweeper game project in C.

Run to You – SNU Baja/Formula 1 Team

Apr 2020 – Oct 2021

[Team Leader](#) – *Baja Chassis Team*

- Led the team in designing and manufacturing an optimized chassis for vehicle performance, and participated in a national student automobile competition(KSAE).

Republic of Korea Army

Mar 2022 – Sep 2023

Squad Leader – *Sergeant, 21st Aviation Brigade*

- Led a 7-soldier squad while supporting maintenance and operation of utility helicopters, ensuring unit readiness and participation in multiple field training exercises.

DB Business Management Competition

Jan 2024

[1st Prize](#) – *Business Performance*

- Gained experience in corporate management (inventory, HR, finance) and achieved top performance in a business simulation competition.

Scholarships

Merit-based Scholarship

Department of Mechanical Engineering

Spring 2025

Seoul National University

Skills

Programming: Python, Matlab, C/C++, PyTorch

Software & Hardware: CAD (SolidWorks, Blender), ANSYS, Arduino, Raspberry Pi, 3D Printing, Laser Cutting

Languages: Korean (Native), English (Fluent, TOEFL 107)